

AdPlatform/DAADD Technical Specification

1. Overview

AdPlatform/DAADD (Data-driven Ad Analytics & Decision Dashboard) is a two-sided AdTech platform for managing, analyzing, and optimizing advertising campaigns across multiple digital channels.

Users:

- **Advertisers:** Create campaigns, connect platform accounts, monitor metrics, receive AI-driven optimization recommendations
- **Consumers/Publishers:** View available ads, claim rewards via QR code redemption

Core Value Proposition:

- Unified metrics aggregation across Google Ads, Meta, TikTok, LinkedIn, Pinterest
 - AI-powered optimization recommendations (budget allocation, underperformer detection, winner scaling)
 - First-party CDP for audience segmentation and activation
 - Creative A/B testing with automatic winner detection
 - Real-time anomaly detection with auto-pause capabilities
 - Geo-targeted campaign management with multi-language support
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2. Architecture (high-level)

AdPlatform/DAADD Platform

- Backend (Express.js + Node.js + MongoDB)
 - Services (21 total)
 - Campaign management
 - Analytics aggregation (unified dashboard)
 - AI optimization
 - Anomaly detection
 - CDP (first-party data)
 - Platform OAuth + token management
 - Webhook event dispatch
 - [... 14 more services]
 - Queues (Bull + Redis)
 - Campaign lifecycle automation
 - Token refresh (hourly)
 - Anomaly checks (async)
 - Report generation
 - Platform Adapters (metrics + audience export)
 - Google Ads
 - Meta
 - TikTok
 - LinkedIn
 - Pinterest
 - Auth (JWT + OTP verification)
- Frontend (React + Vite + React Router)
 - Pages (23 total)
 - Dashboard (campaigns, analytics, AI optimization)
 - Platform accounts settings
 - CDP audience management
 - Reports & exports
 - [... 18 more pages]
 - State (Zustand + TanStack Query v5)
 - Components (atomic design)
 - Hooks (custom data fetching)
- Mobile (Expo + React Native)
 - Consumer-facing (view ads, claim rewards)
 - Auth flow
 - Ad detail + reward tracking
- Infrastructure
 - MongoDB (campaigns, users, events)
 - Redis (cache, queues, rate limiting)
 - Email (Resend API)
 - External platform APIs (Google, Meta, TikTok, LinkedIn, Pinterest)

3. Product Requirements

3.1 Core Campaigns Module

- Create, read, update, delete campaigns
- Campaign status machine: DRAFT → ACTIVE → COMPLETED/PAUSED
- Budget tracking (spent vs. total)
- Date-based lifecycle (start_date, end_date)
- AI optimization toggle (on/off per campaign)
- Creative attachment (multiple creatives per campaign)
- Campaign cloning (duplicate with reset budget)
- A/B testing (assign creatives to variants, auto-detect winner after 10K impressions)
- Multi-language support (14 languages)

3.2 Analytics Dashboard

- Unified metrics aggregation (all platforms combined)
- Per-platform breakdown (Google, Meta, TikTok, LinkedIn, Pinterest)
- Per-campaign breakdown
- Metrics: impressions, clicks, spend, conversions, CTR, CPC, CPA, ROAS
- Date range filtering
- CSV/PDF export
- Funnel analysis (impression → click → conversion)
- Device & geographic heatmaps (100-view gate)

3.3 AI-Driven Optimization

- Auto-detect high-ROAS platforms, suggest budget reallocation
- Detect underperformers, recommend pause/optimization
- Scale winning campaigns (>3x ROAS)
- Platform-specific adjustments (targeting, bidding, creative)
- Recommendation impact scoring (0-100% expected ROAS improvement)
- Auto-apply with audit trail
- Manual dismiss capability

3.4 First-Party Data Platform (CDP)

Phase 1 — Data Collection:

- Pixel endpoint for conversion tracking (public, rate-limited)
- CSV import for email lists
- Cross-device user unification (deterministic hashing)
- User profile enrichment (behavioral metrics, segments)

Phase 2 — Audience Activation:

- Segment audiences with rule-based targeting (field/operator/value)
- Export to platform custom audiences (hashed email/phone)
- Activation tracking per platform
- Sync status monitoring

Phase 3 — CRM Integrations:

- Connect Shopify, Salesforce, Klaviyo
- Sync customer data (order history, revenue, recency)
- Predictive scoring (churn risk, LTV, engagement, conversion probability)
- Lookalike audience generation (similarity-based)

3.5 Platform Integrations

OAuth Flow:

- Authorization URL generation per platform
- Callback handling + token exchange
- Token encryption (AES-256-GCM) at rest
- Automatic refresh (hourly check, refresh 24h before expiry)

Metrics Syncing:

- Fetch last 30 days of metrics from each platform
- Aggregate into unified view
- Manual sync trigger (/dashboard/sync-now)
- Configurable sync frequency (realtime, hourly, daily)

Supported Platforms:

- Google Ads (API v15)
- Meta (Graph API v18)
- TikTok (Business API)
- LinkedIn (Campaign Manager v2)
- Pinterest (Ads API v5)

3.6 Anomaly Detection

Auto-trigger on:

- CTR drops >20% vs 7-day average
- CPA spikes >25% vs 7-day average
- Spend anomalies (>30% variance)
- Conversion rate collapse

Actions:

- Auto-pause campaign (optional)
- Notify advertiser (email + in-app)
- Webhook dispatch
- Audit log entry

3.7 Rewards & Redemption

- Advertiser-issued reward credits
- QR code generation (HMAC-SHA256 signed, 2-min TTL)
- One-time redemption per QR code
- FIFO balance deduction
- Merchant redemption workflow (approve/reject)

3.8 Team Collaboration

- Role-based access (admin, campaign_manager, viewer)
- Invitation workflow
- Audit logging of all actions
- Permissions enforcement on sensitive operations

4. Technical Stack

Layer	Technology	Version
Backend	Node.js + Express.js	20.x / 7.x
Backend DI	tsyringe	4.x
Database	MongoDB + Mongoose	8.x
Cache	Redis	7.x
Task Queue	Bull	4.x
Auth	JWT (jsonwebtoken)	9.x
Email	Resend API	v1
Frontend	React + Vite	18.x / 5.x
Frontend Router	React Router	6.x
State	Zustand + TanStack Query	4.x / 5.x
Mobile	Expo + React Native	SDK 55
Validation	Zod	3.x

5. API Endpoints (RESTful)

Base: /api/v1

Feature	Endpoint	Auth	Methods
Auth	/auth/register, /auth/login, /auth/refresh	None	POST
Campaigns	/campaigns	JWT	GET, POST, PATCH, DELETE
Creatives	/campaigns/:id/creatives	JWT	GET, POST
Analytics	/analytics/dashboard, /analytics/:campaignId	JWT	GET
Heatmaps	/heatmaps/:campaignId	JWT	GET
AI Optimization	/ai/recommendations, /ai/apply/:recommendationId	JWT	GET, POST
Anomalies	/anomalies/:campaignId	JWT	GET
Conversion Pixel	/pixel/:campaignId	None (public)	POST
Platform Accounts	/platform-accounts	JWT	GET, POST, PATCH, DELETE
OAuth	/oauth/authorize/:platform, /oauth/callback/:platform	Token-aware	GET
CDP (Data)	/cdp/pixel, /cdp/import-csv, /cdp/profiles	JWT	POST, GET
CDP (Audiences)	/cdp/audiences	JWT	GET, POST
CRM Integration	/cdp-advanced/crm/integrate, /cdp-advanced/crm/sync	JWT	POST, GET

Webhooks	/webhooks	JWT	GET, POST, DELETE
Unified Dashboard	/dashboard/unified-metrics, /dashboard/sync-now	JWT	GET, POST
Rewards	/rewards/balance, /rewards/claim	JWT	GET, POST
Redemption	/redemption/generate-qr, /redemption/approve	JWT	POST

6. Database Schema (core entities)

- **User** — advertiser + admin accounts
 - **Campaign** — advertising campaigns
 - **Creative** — ad creatives (images, copy)
 - **AdEvent** — impressions, clicks, conversions, viewability
 - **PlatformAccount** — OAuth tokens for Google, Meta, TikTok, LinkedIn, Pinterest
 - **UserProfile** — CDP unified user records
 - **Audience** — segmented user lists
 - **AudienceActivation** — per-platform audience sync status
 - **Reward** — advertiser-issued credits
 - **Redemption** — QR code redemptions
 - **Review** — user reviews of campaigns
 - **TeamMember** — RBAC assignments
 - **AnomalyAlert** — detected anomalies + notifications
 - **AiAuditLog** — recommendation application history
 - **Webhook** — advertiser-configured event webhooks
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7. Quality Standards

Per **PROMPT_PLAYBOOK.md**:

7.1 Commits

- **Format:** Conventional Commits (feat, fix, chore, docs, test, refactor, perf, build, ci)
- **Body:** Detailed — explain WHY, not WHAT
- **No AI authorship:** Never include "Built with Claude" or similar in messages
- **Atomic:** One concern per commit
- **Frequency:** Small, logically-grouped changes

7.2 Tests

- **Unit tests:** Mock external dependencies, use Jest
- **Integration tests:** Real database/Redis via testcontainers (when applicable)
- **End-to-end:** Smoke tests for new API endpoints
- **Coverage:** Aim for >75% on critical paths (auth, payment, data mutation)

7.3 Documentation

- **Doc comments:** Every exported function, class, interface, constant
- **Architecture doc:** ARCHITECTURE.md — live code map, updated per commit
- **Contributing guide:** CONTRIBUTING.md — workflow, testing, deployment
- **Specification:** This file — product + tech decisions

7.4 Design Patterns

- **Dependency Injection (tsyringe):** All services injectable, composition-root in container.ts
- **Repository pattern:** All DB access goes through repository interfaces
- **Adapter pattern:** Platform integrations (OAuth, metrics, audience export)
- **Strategy pattern:** Swappable storage providers (Cloudinary, S3, local)
- **Provider pattern:** External services (email, AI) behind interfaces with stub + real impl

7.5 Linting

- **ESLint:** Configured in backend/.eslintrc.json and frontend/.eslintrc.json
 - **Prettier:** Format enforcement
 - **TypeScript:** strict: true
 - **Pre-commit hooks:** Lint + format check before commit (configured in .husky/pre-commit)
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8. Deployment

8.1 Environments

Env	Backend	Frontend	Mobile	Notes
Dev	localhost:4000	localhost:3000	Expo dev server	Local MongoDB, Redis
Staging	Render	Vercel	EAS Build preview	Cloud MongoDB, Redis
Production	Render	Vercel	EAS Build release	Production DBs, stripe mode

8.2 CI/CD

- **GitHub Actions:** Lint + test on PR
- **Pre-merge gate:** All tests + lints must pass
- **Auto-deploy:** Merge to main triggers Vercel (frontend), Render (backend)

8.3 Environment Variables

Backend (.env):

```
NODE_ENV=production
DATABASE_URL=mongodb+srv://...
REDIS_URL=redis://...
JWT_SECRET=<strong-random>
ENCRYPTION_SECRET=<strong-random>
RESEND_API_KEY=<from Resend>
GOOGLE_OAUTH_CLIENT_ID=<from Google Cloud>
GOOGLE_OAUTH_CLIENT_SECRET=<...>
META_OAUTH_APP_ID=<from Meta Developer>
META_OAUTH_APP_SECRET=<...>
TIKTOK_OAUTH_CLIENT_ID=<from TikTok>
TIKTOK_OAUTH_CLIENT_SECRET=<...>
LINKEDIN_OAUTH_CLIENT_ID=<from LinkedIn>
LINKEDIN_OAUTH_CLIENT_SECRET=<...>
PINTEREST_OAUTH_CLIENT_ID=<from Pinterest>
PINTEREST_OAUTH_CLIENT_SECRET=<...>
CORS_ORIGINS=https://app.adplatform.com,https://staging.adplatform.com
BACKEND_URL=https://api.adplatform.com
FRONTEND_URL=https://app.adplatform.com
STRIPE_SECRET_KEY=<if using Stripe>
ANTHROPIC_API_KEY=<for AI Creative generation>
```

Frontend (.env):

```
VITE_API_URL=https://api.adplatform.com/api/v1
VITE_APP_NAME=AdPlatform
```

9. Non-Functional Requirements

- **Performance:** API responses <200ms p95, dashboard load <1.5s (with 500+ campaigns)
- **Availability:** 99.5% uptime SLA
- **Scalability:** Support 10K advertisers, 1M campaigns, 10B events
- **Security:** OAuth 2.0, JWT with 1-hour expiry, password hashing (bcrypt), encrypted tokens at rest
- **Compliance:** GDPR right-to-be-forgotten, CCPA compliance, no PII in logs
- **Observability:** Structured logging, error tracking (Sentry), monitoring (DataDog)

10. Known Constraints & Decisions

1. **Conversion value:** Assumed \$50 per conversion for ROAS calculation (configurable per advertiser in future)
2. **Attribution window:** 30-day lookback for cohort analysis (configurable per campaign)
3. **A/B test winner:** Auto-declared after 10K impressions or 7 days, whichever comes first
4. **QR code TTL:** 2 minutes (security trade-off vs. convenience)
5. **Anomaly detection:** Triggered on 7-day moving average baseline (not hard thresholds)
6. **OAuth token storage:** Encrypted with AES-256-GCM, decrypted only when making API calls
7. **Email delivery:** Resend API (replaced legacy Nodemailer setup for reliability)

11. Open Questions (None — all decided via PROMPT_PLAYBOOK decision-making)

12. Future Roadmap

- **Phase 2:** Real-time WebSocket metrics streaming
 - **Phase 3:** Programmatic RTB bidding
 - **Phase 4:** Connected TV (CTV) ad management
 - **Phase 5:** Audio advertising (Spotify, Podcast APIs)
 - **Phase 6:** Retail Media Networks (Amazon, Walmart, Target)
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Last Updated: May 17, 2026

Maintained By: Development Team

Status: Active